



**Wydział Matematyki
i Nauk Informatycznych**

POLITECHNIKA WARSZAWSKA

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Convolutional neural networks

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Team members



Maciej Momot



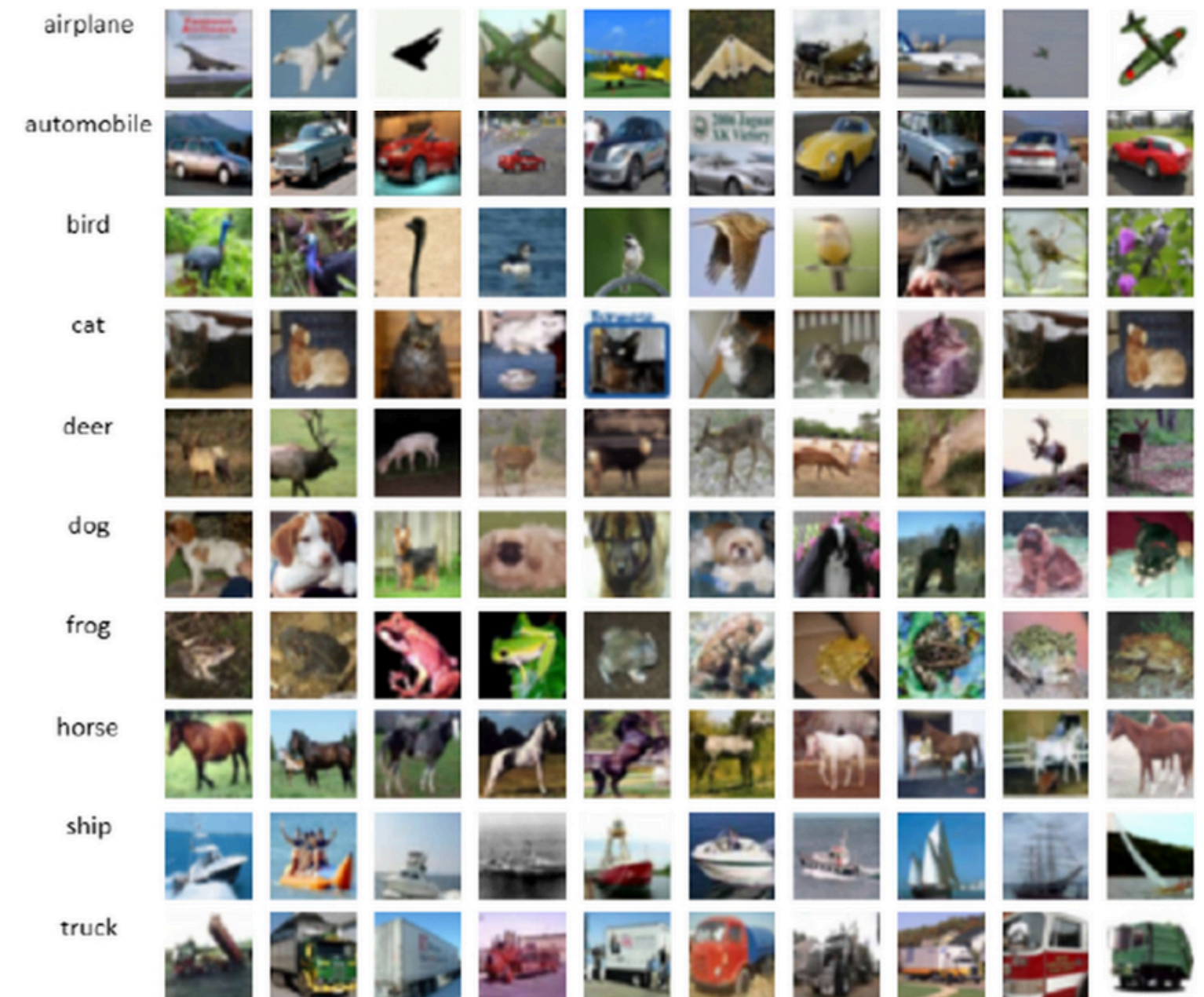
Mikołaj Rowicki



Jakub Półtorak

Agenda

1. Selected Architectures
2. Experimental Strategy
3. Training Hyperparameters
4. Regularization Techniques
5. Classical Data Augmentation
6. Advanced Data Augmentation
7. Few-Shot Learning
8. Model Explainability (XAI)
9. Architecture Comparison & Ensembling

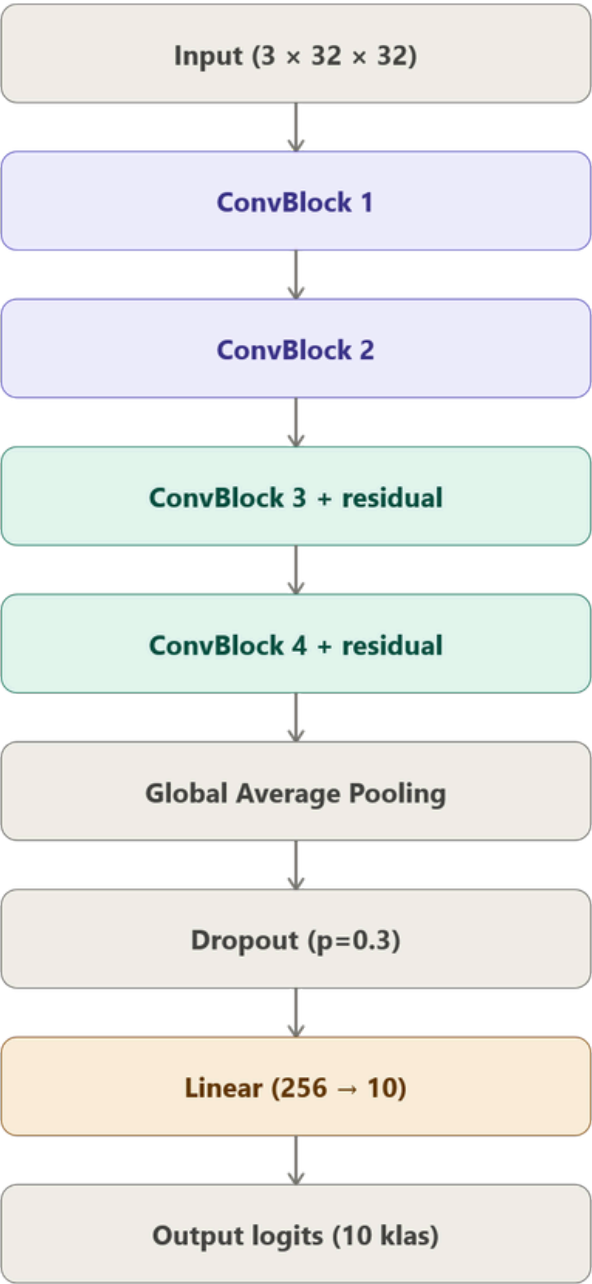


Sample views of the classes from CIFAR-10

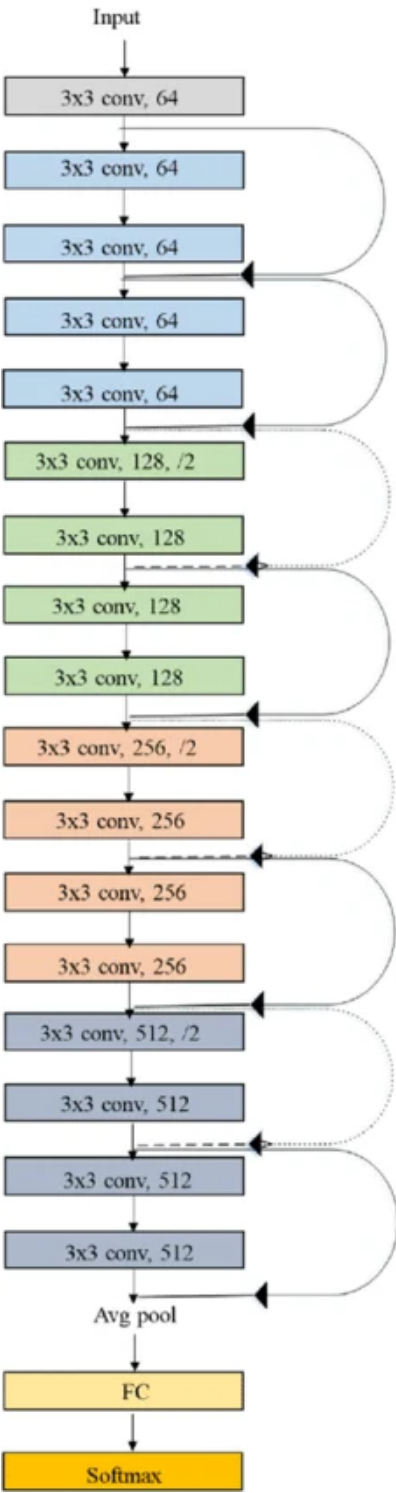
Architectures

Custom

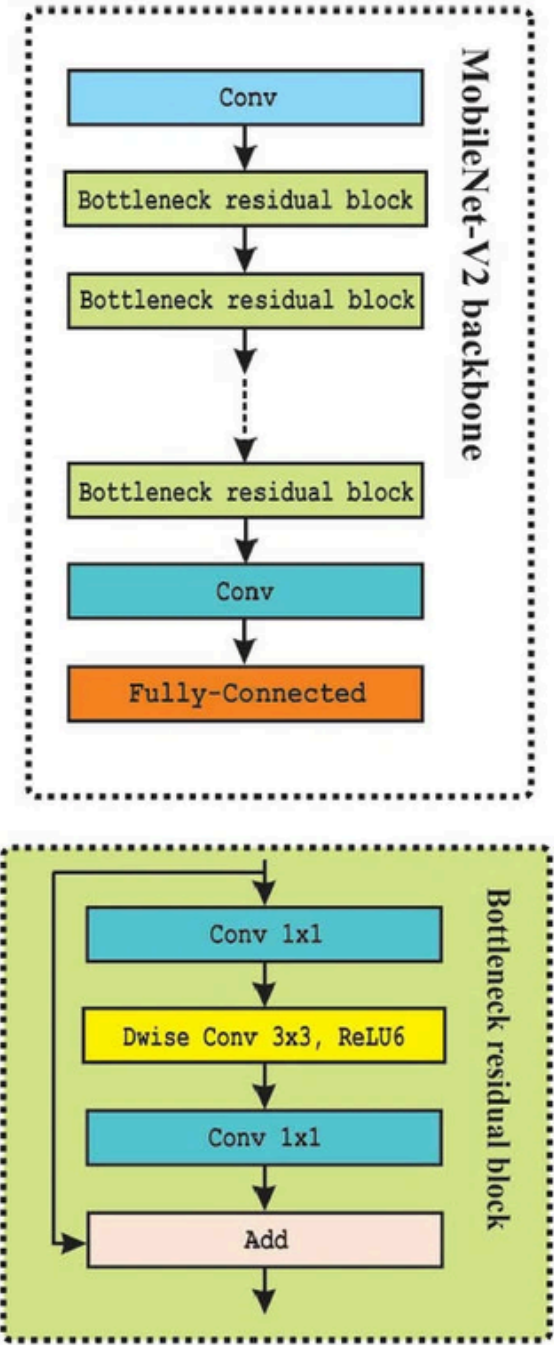
- Conv → BN → ReLU → Conv → BN → ReLU → MaxPool
- Conv → BN → ReLU → Conv → BN → (+skip) → ReLU → MaxPool



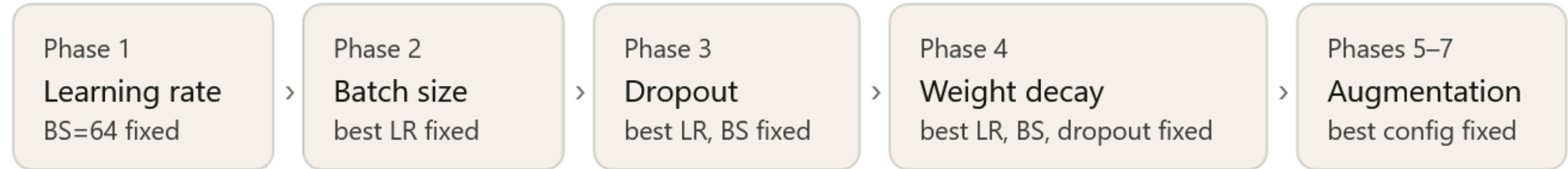
Resnet18



MobileNetV2



Experimental strategy



Hyperparameter	Values tested	Baseline
Learning rate	10^{-2} , 10^{-3} , 10^{-4}	10^{-3}
Batch size	32, 64, 128	64
Dropout rate	0.0, 0.2, 0.3	0.0
Weight decay	0.0, 10^{-5} , 10^{-4}	0.0

AUGMENTATION TECHNIQUES

Baseline
no augmentation

Horizontal flip
 $p = 1.0$

Gaussian blur
kernel 3×3 , $\sigma \in [0.1, 2.0]$

Color jitter
brightness, contrast, sat., hue

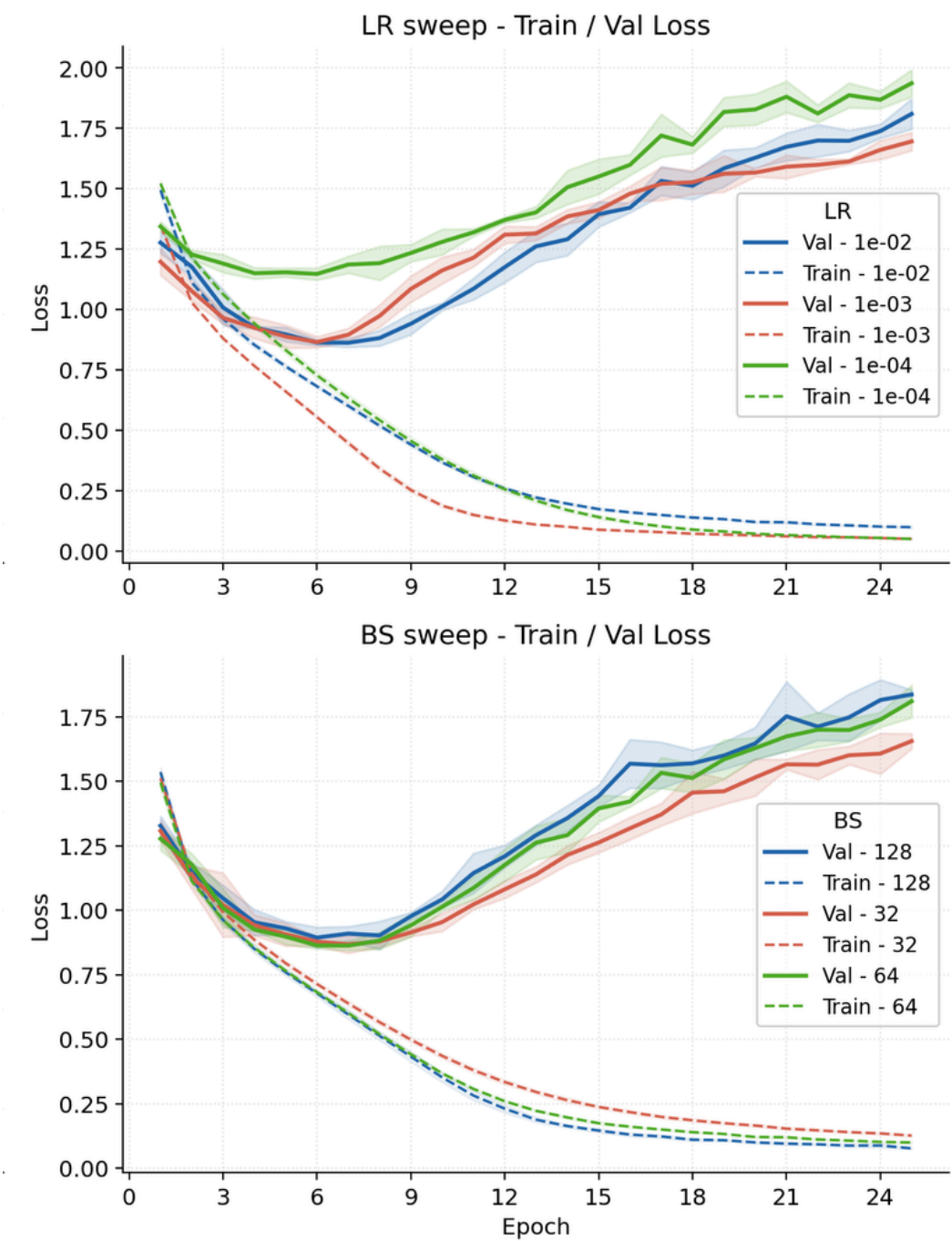
CutOut
 16×16 px, $p = 1.0$

CutMix $\alpha=1$ / $\alpha=4$
 $p = 0.5$ per batch

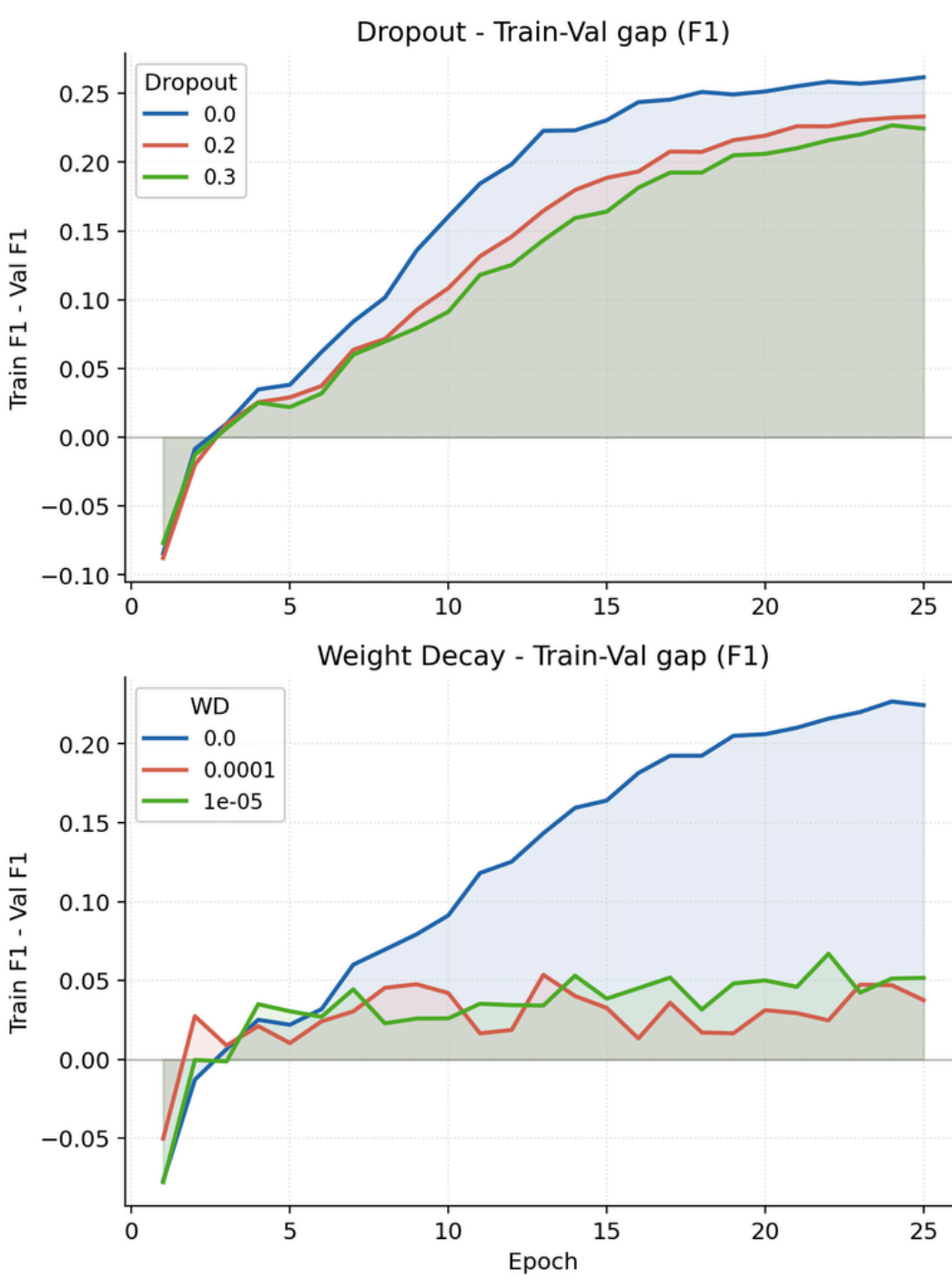
Random combined
uniform random choice from 1 of 4 techniques

Custom CNN - training hyperparameters

Phase 1-2: Effect of Learning Rate and Batch Size



Phase 3-4: Regularization - Dropout and Weight Decay



Learning Rate	Val F1 (mean $\pm$ std)	Test F1 (mean $\pm$ std)
10^{-2}	0.7144 ± 0.0023	0.7124 ± 0.0024
10^{-3}	0.7106 ± 0.0024	0.7076 ± 0.0024
10^{-4}	0.6040 ± 0.0083	0.6018 ± 0.0084

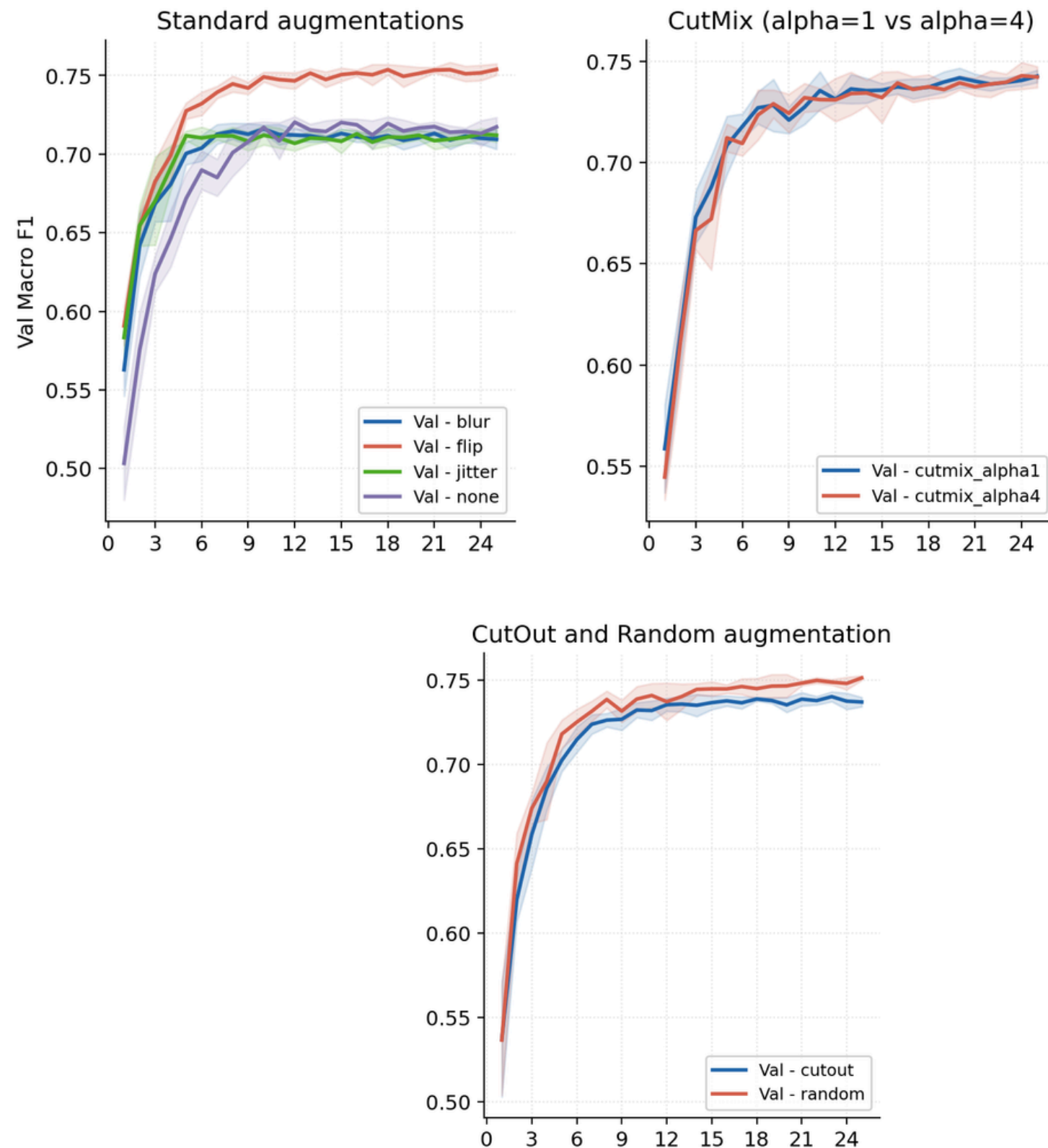
Batch Size	Val F1 (mean $\pm$ std)	Test F1 (mean $\pm$ std)
32	0.7144 ± 0.0020	0.7114 ± 0.0031
64	0.7144 ± 0.0023	0.7124 ± 0.0024
128	0.7122 ± 0.0019	0.7103 ± 0.0034

Dropout	Val F1 (mean $\pm$ std)	Test F1 (mean $\pm$ std)
0.0	0.7144 ± 0.0023	0.7124 ± 0.0032
0.2	0.7226 ± 0.0025	0.7205 ± 0.0026
0.3	0.7233 ± 0.0017	0.7214 ± 0.0025

Weight Decay	Val F1 (mean $\pm$ std)	Test F1 (mean $\pm$ std)
0.0	0.7233 ± 0.0017	0.7214 ± 0.0025
10^{-5}	0.7116 ± 0.0032	0.7092 ± 0.0025
10^{-4}	0.6317 ± 0.0070	0.6301 ± 0.0073

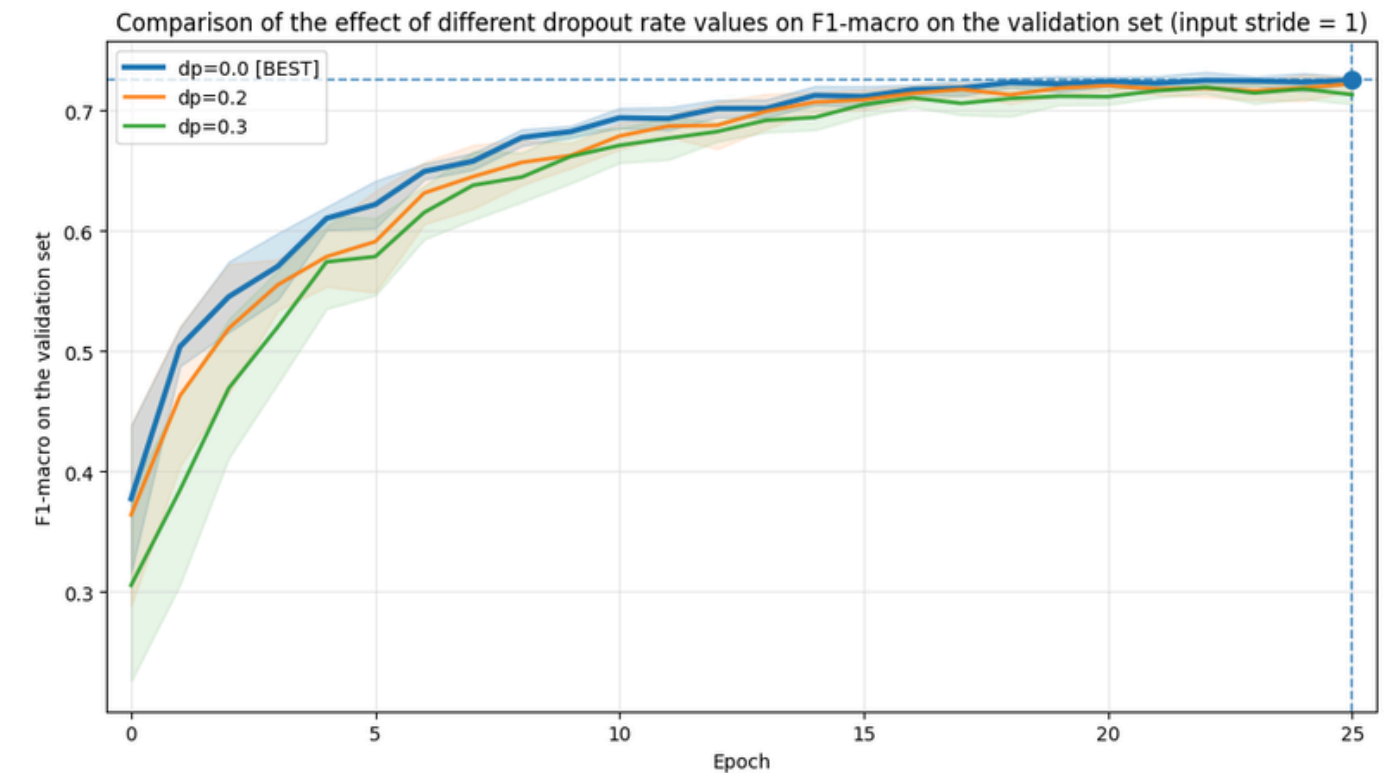
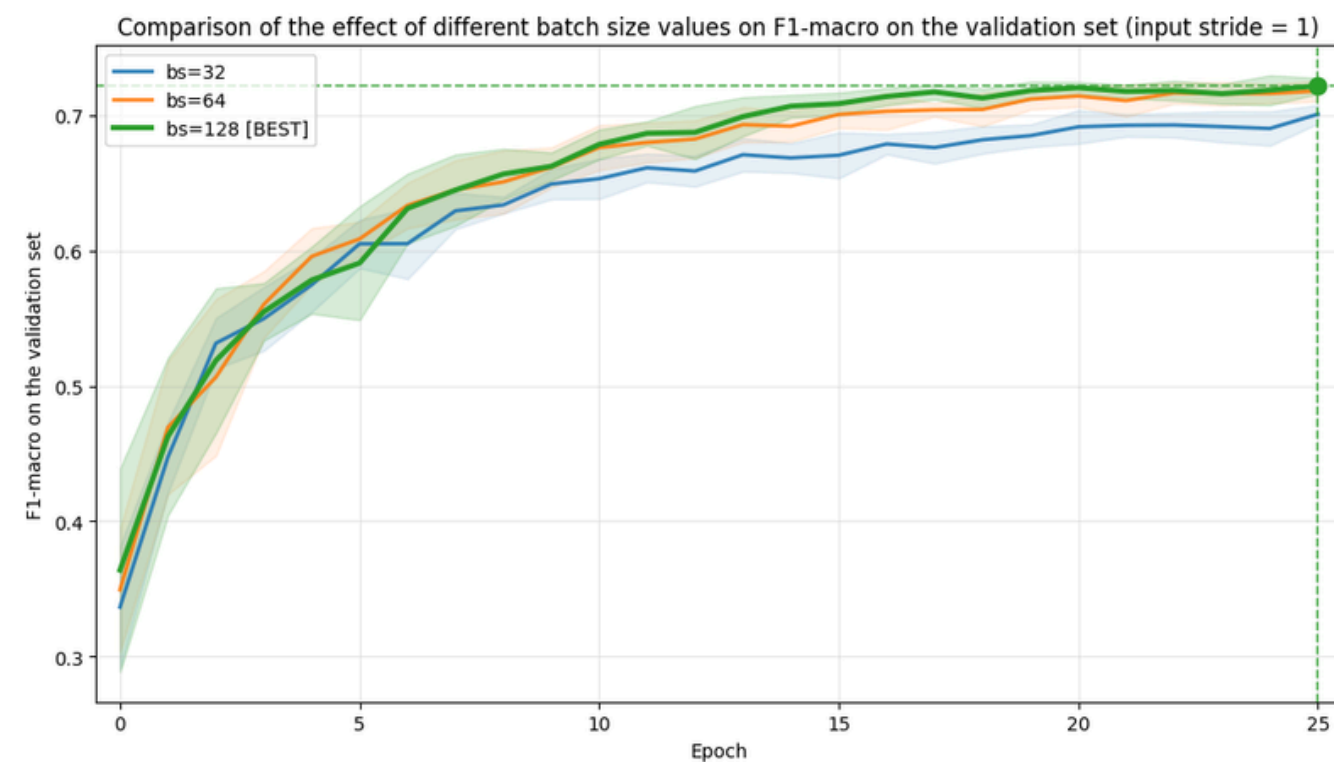
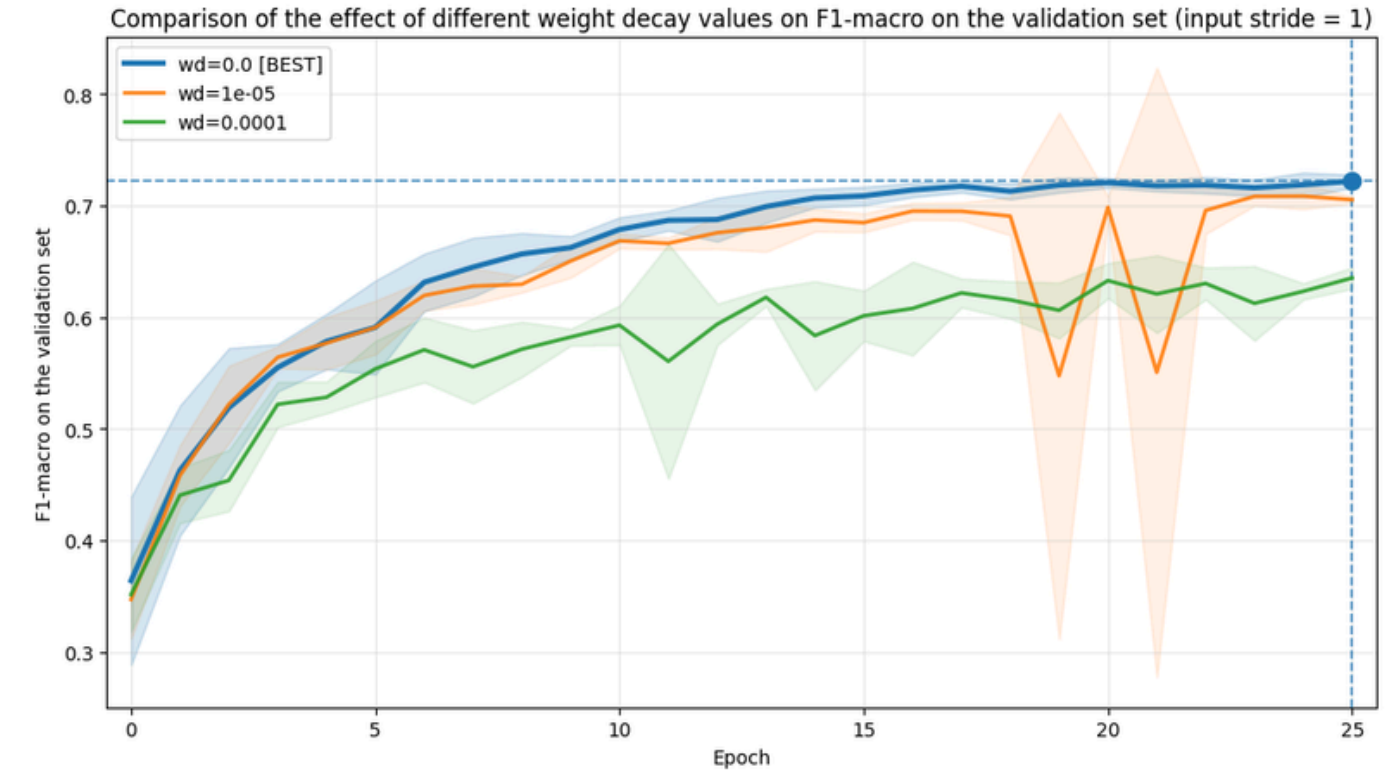
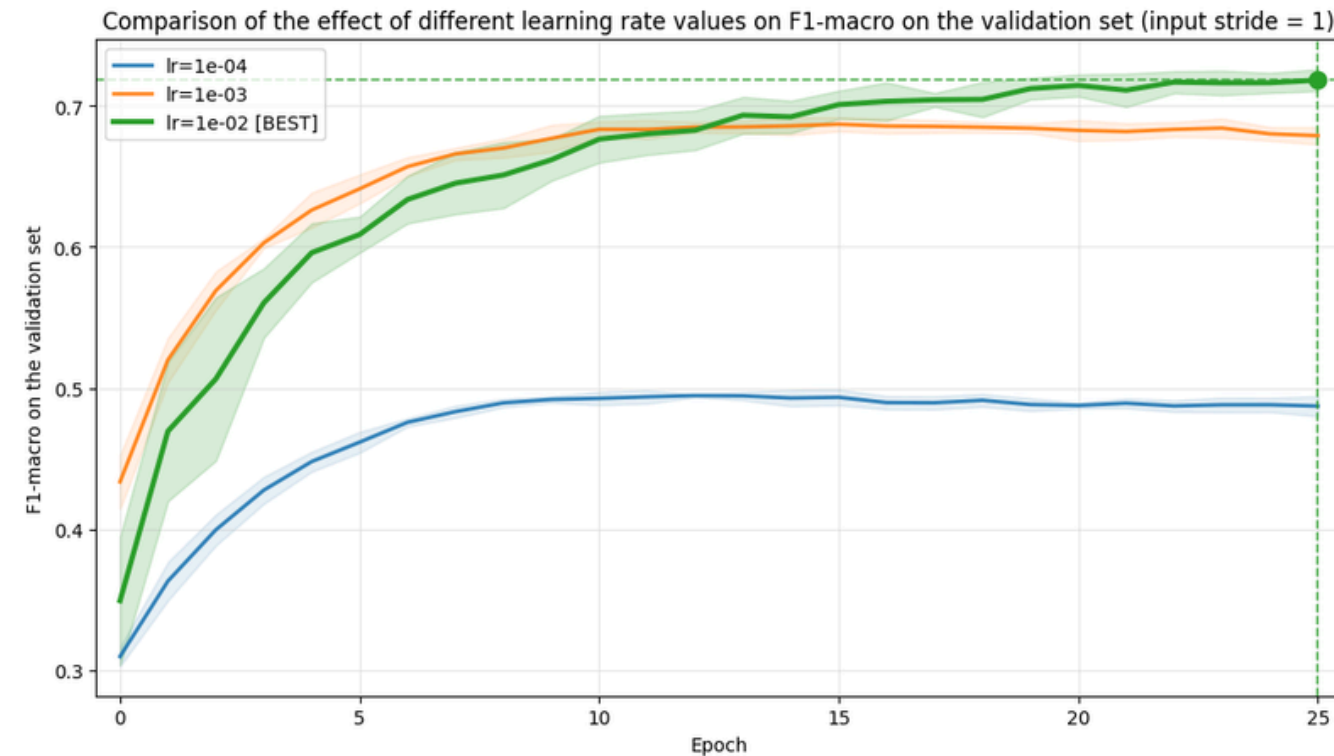
Custom CNN - augmentation

Phase 5-7: Data Augmentation Techniques

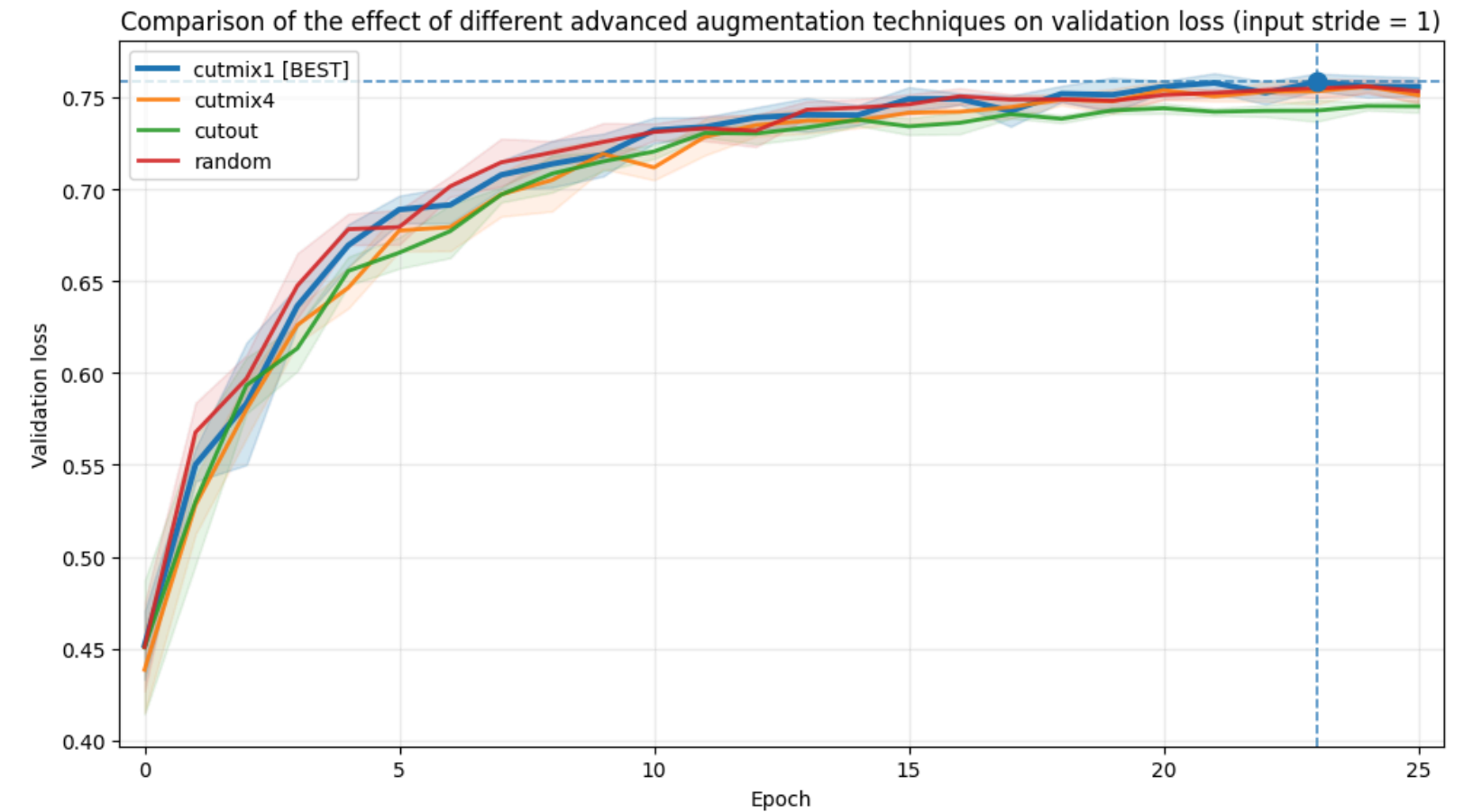
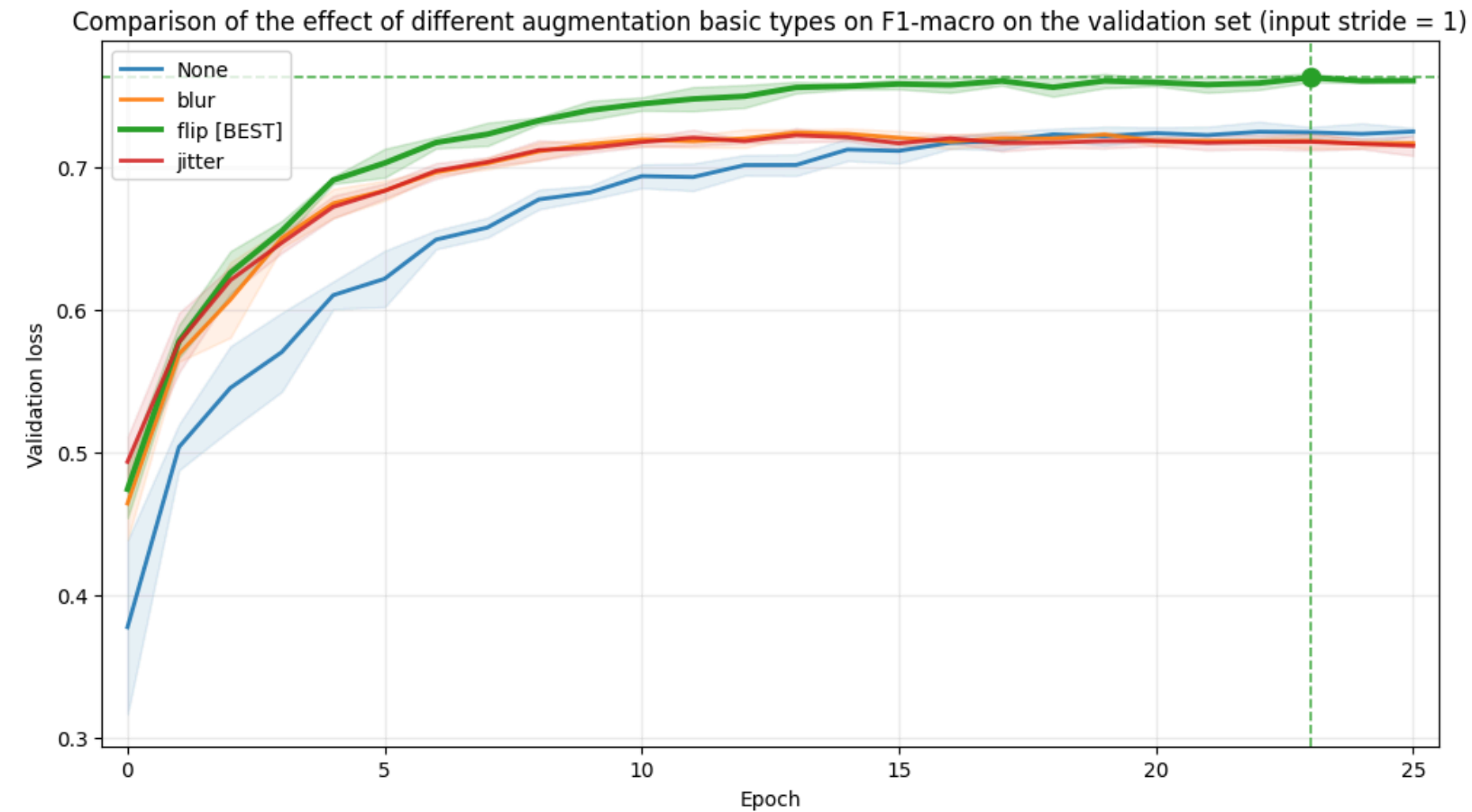


Category	Augmentation	Test F1 (mean $\pm$ std)
Standard	None	0.7214 ± 0.0025
	Flip	0.7535 ± 0.0023
	Blur	0.7158 ± 0.0021
	Jitter	0.7144 ± 0.0024
Region-based	CutMix $\alpha = 1$	0.7423 ± 0.0020
	CutMix $\alpha = 4$	0.7432 ± 0.0029
	CutOut	0.7389 ± 0.0010
Composite	Random	0.7498 ± 0.0010

Mobilenet - training hyperparameters



Mobilenet - augmentation

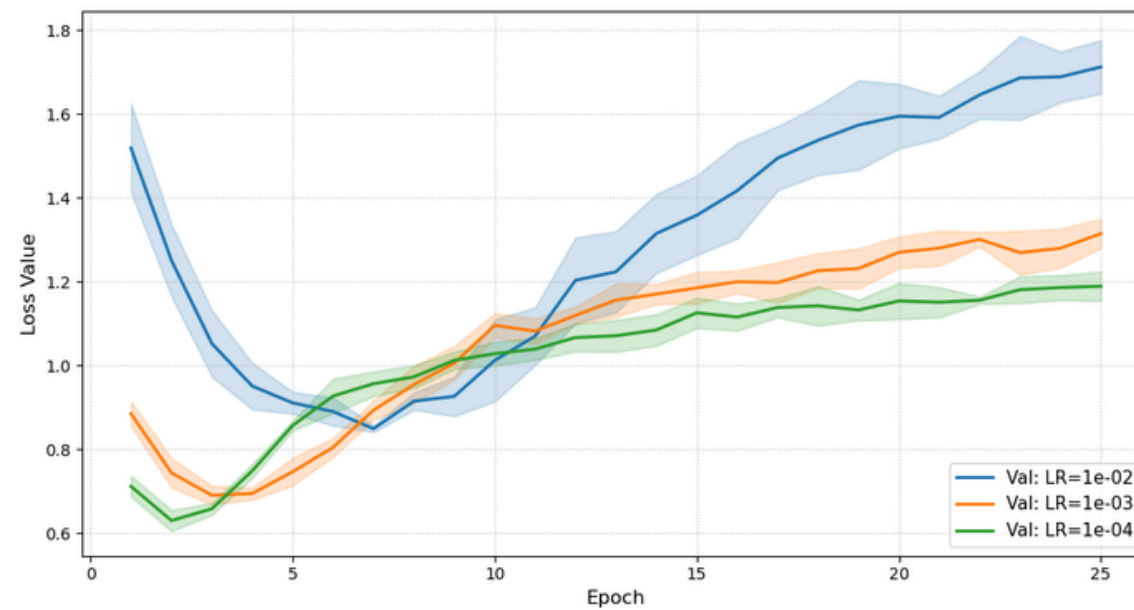


Augmentation Type	Validation F1(mean ± std)	Test F1(mean ± std)
None	0.7287 ± 0.0046	0.7219 ± 0.0033
blur	0.7262 ± 0.0011	0.7223 ± 0.0012
flip	0.7640 ± 0.0018	0.7602 ± 0.0016
jitter	0.7248 ± 0.0017	0.7222 ± 0.0015

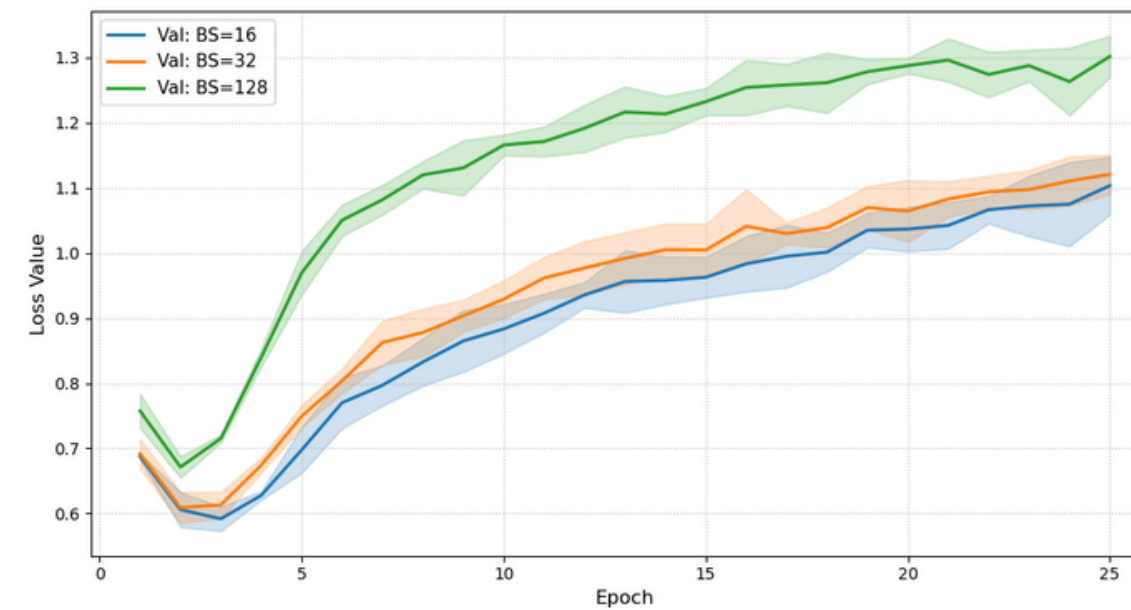
Augmentation Type	Validation F1(mean ± std)	Test F1(mean ± std)
cutmix($\alpha = 1$)	0.7614 ± 0.0015	0.7589 ± 0.0012
cutmix($\alpha = 4$)	0.7580 ± 0.0035	0.7554 ± 0.0035
cutout	0.7477 ± 0.0020	0.7456 ± 0.0024
random	0.7574 ± 0.0036	0.7541 ± 0.0037

Resnet - training hyperparameters

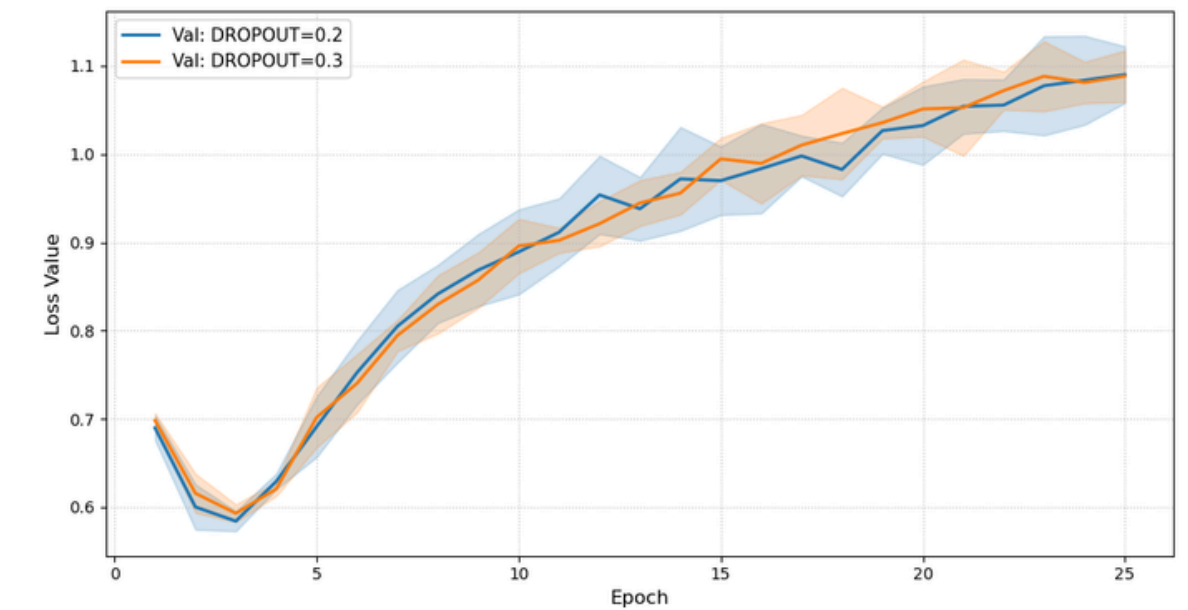
Hyperparameter Comparison: LR
(Fixed parameters: BS=64)



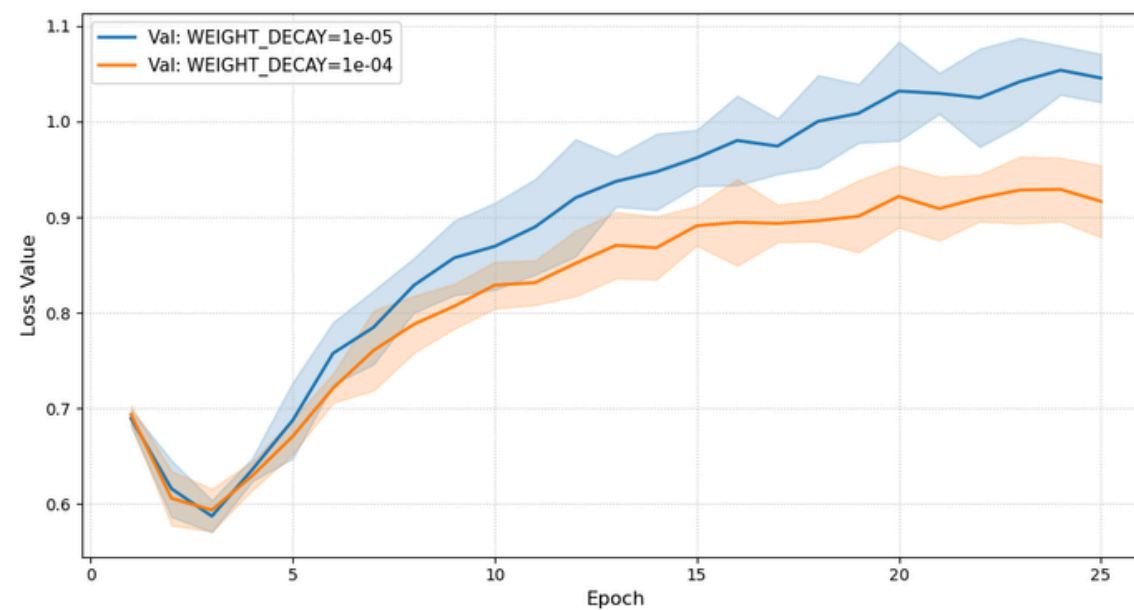
Hyperparameter Comparison: BS
(Fixed parameters: LR=1e-04)



Hyperparameter Comparison: DROPOUT
(Fixed parameters: LR=1e-04_BS=16)



Hyperparameter Comparison: WEIGHT_DECAY
(Fixed parameters: LR=1e-04_BS=16_DROPOUT=0.2)



Weight Decay

Validation F1

Test F1

0.0 (None)

0.8014 ± 0.0063

0.7991 ± 0.0059

10^{-5}

0.8013 ± 0.0060

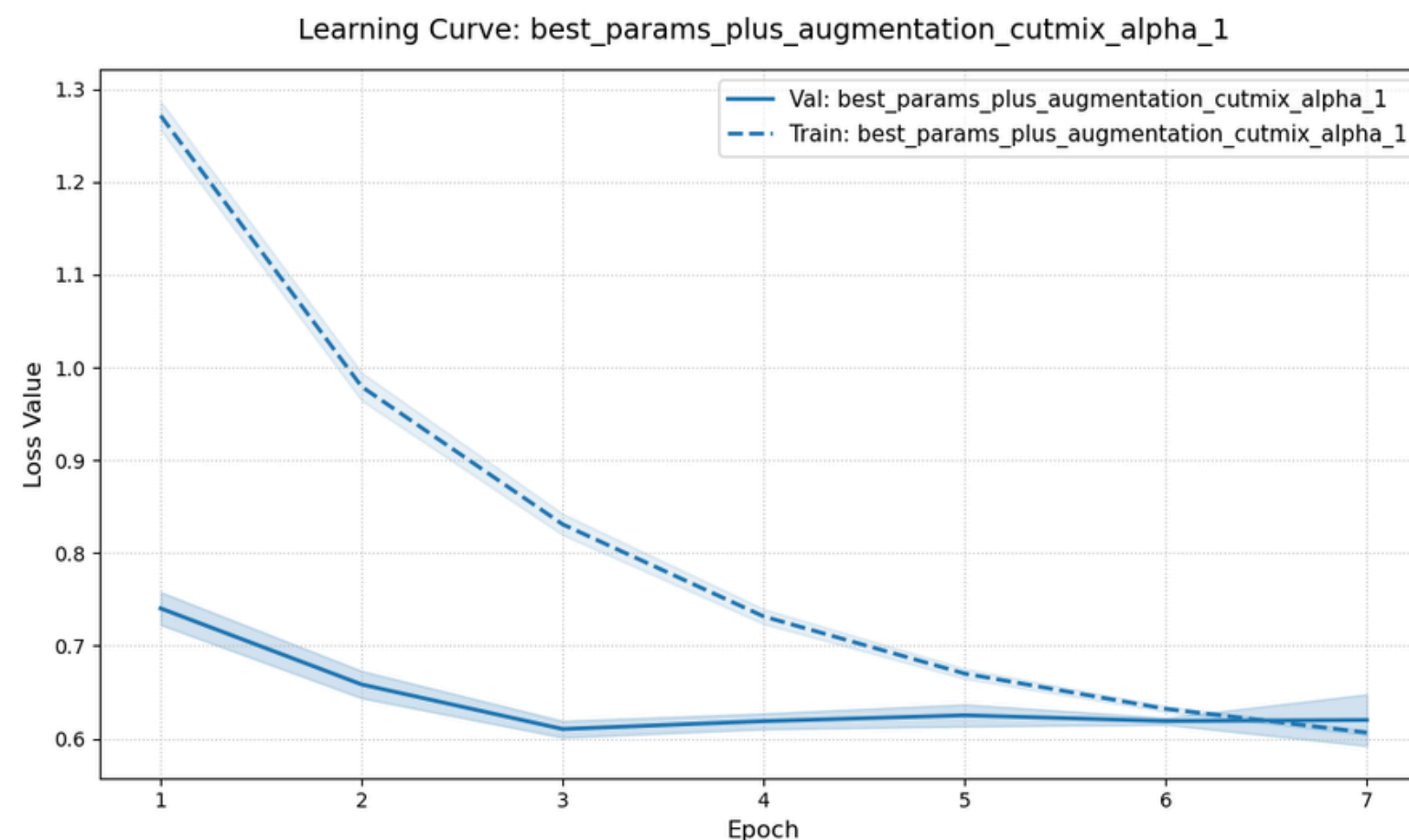
0.7989 ± 0.0069

10^{-4}

0.7998 ± 0.0044

0.7965 ± 0.0051

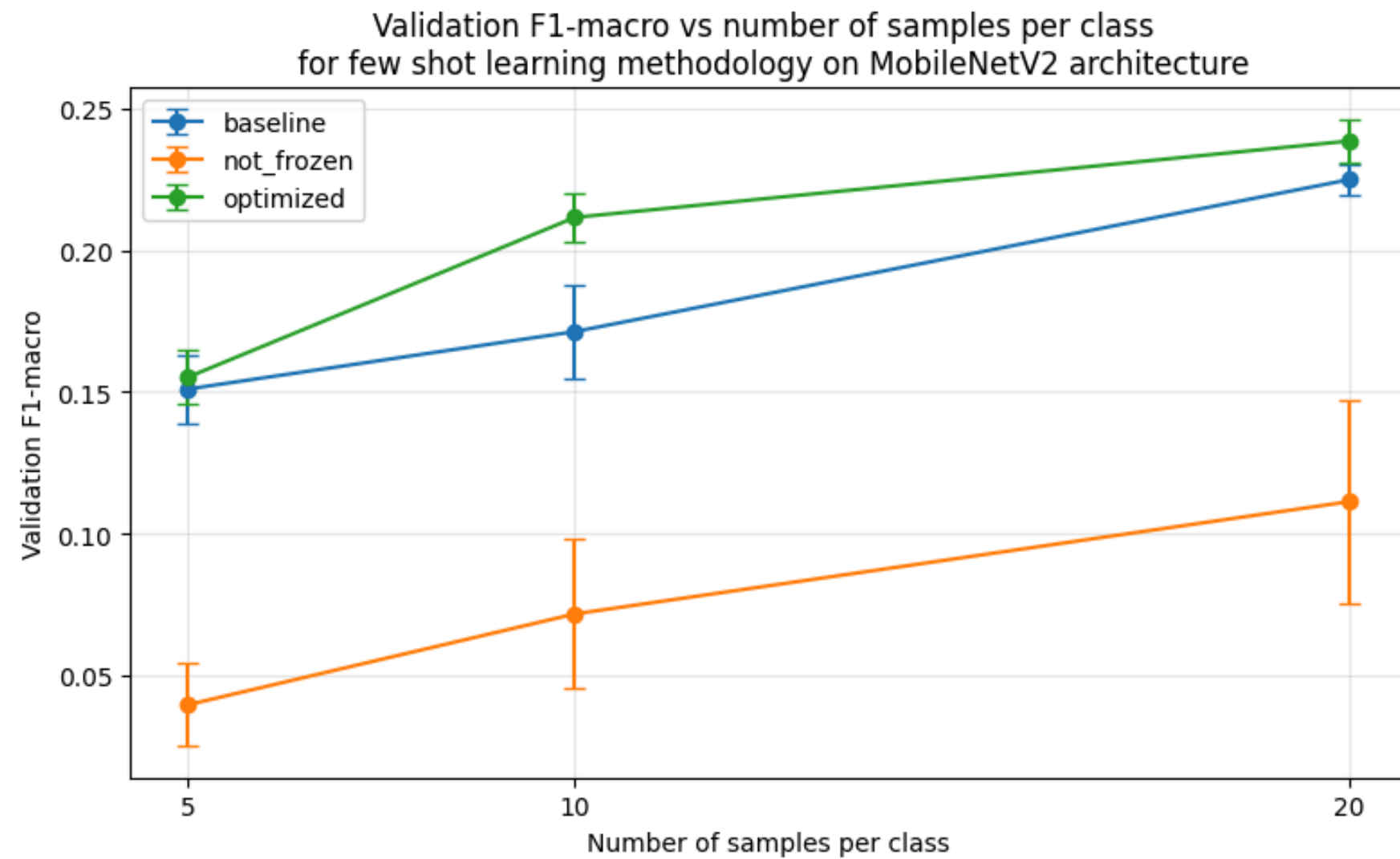
Resnet - augmentation



Augmentation technique	Validation F1	Test F1
Flip	0.8235 ± 0.0010	0.8214 ± 0.0011
Blur	0.7868 ± 0.0024	0.7855 ± 0.0028
Jitter	0.7925 ± 0.0031	0.7901 ± 0.0031
Cutmix ($\alpha = 1$)	0.8241 ± 0.0033	0.8224 ± 0.0028
Cutmix ($\alpha = 4$)	0.8227 ± 0.0025	0.8206 ± 0.0023
Cutout	0.8032 ± 0.0016	0.8011 ± 0.0023
Random	0.8111 ± 0.0043	0.8095 ± 0.0048

Few shot learning

MobileNetV2 results



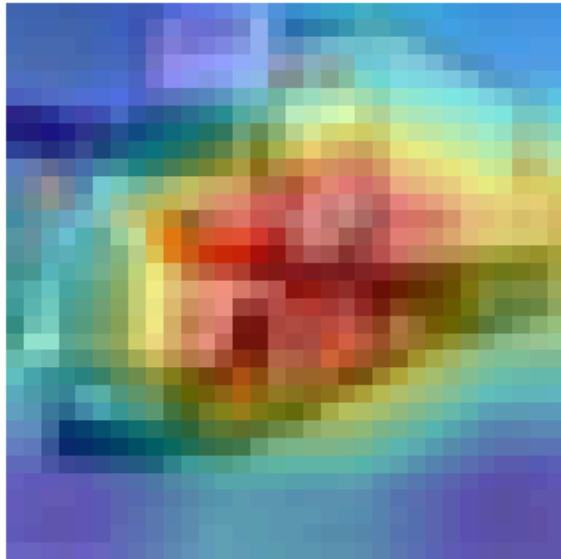
Resnet18 results

Samples per class	Validation F1	Test F1
5	0.1802 ± 0.0538	0.1665 ± 0.0730
10	0.1585 ± 0.0250	0.1716 ± 0.0299
20	0.2083 ± 0.0121	0.2671 ± 0.0241

Model explainability - GradCAM++

Grad-CAM++ Heatmaps at layer: layer4.1.conv2

airplane (idx: 0)



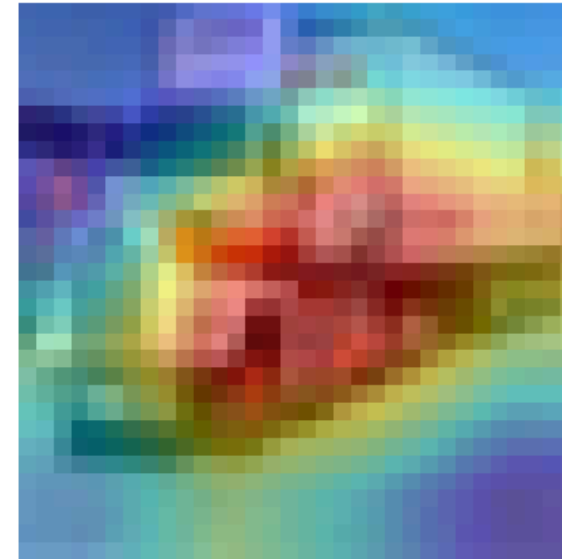
automobile (idx: 1)



bird (idx: 2)



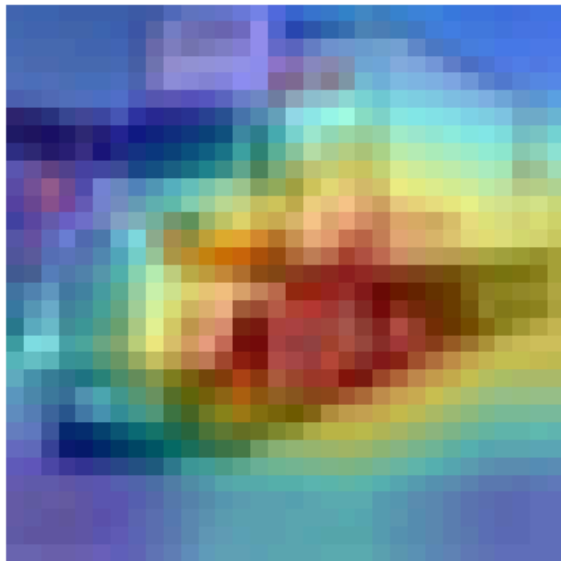
cat (idx: 3)



deer (idx: 4)



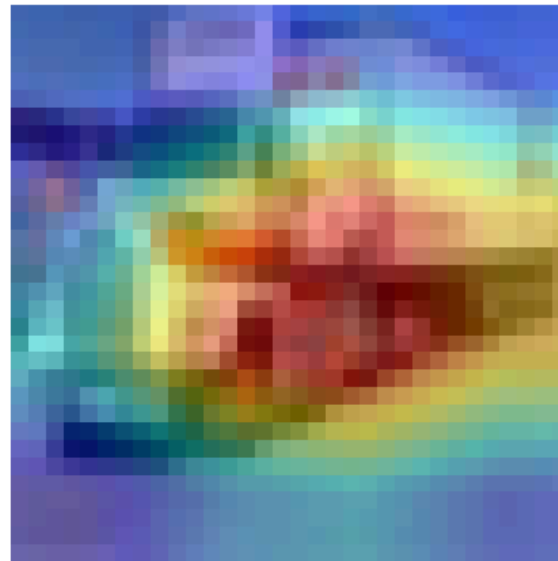
dog (idx: 5)



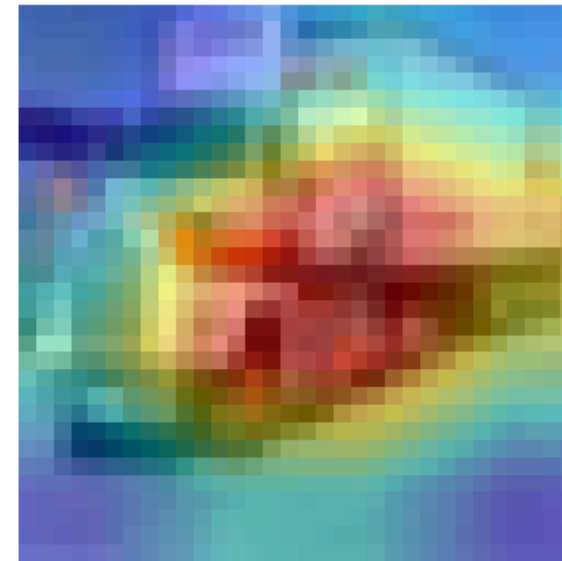
frog (idx: 6)



horse (idx: 7)



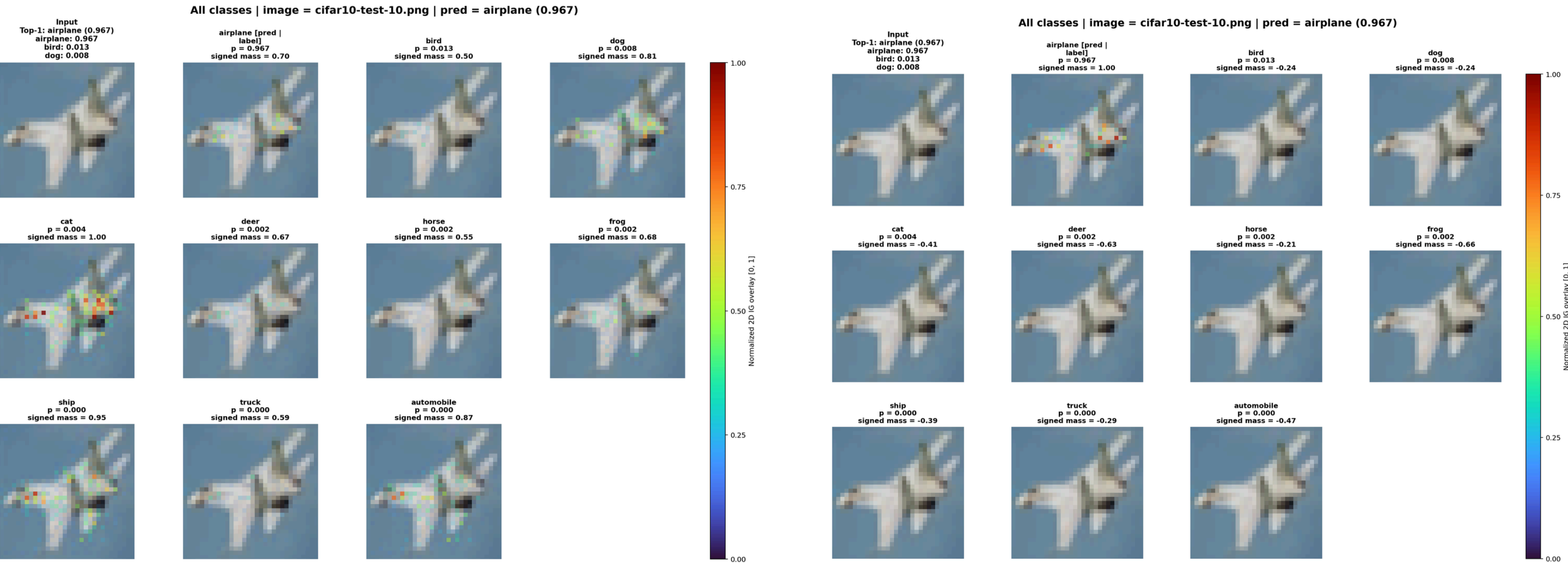
ship (idx: 8)



truck (idx: 9)



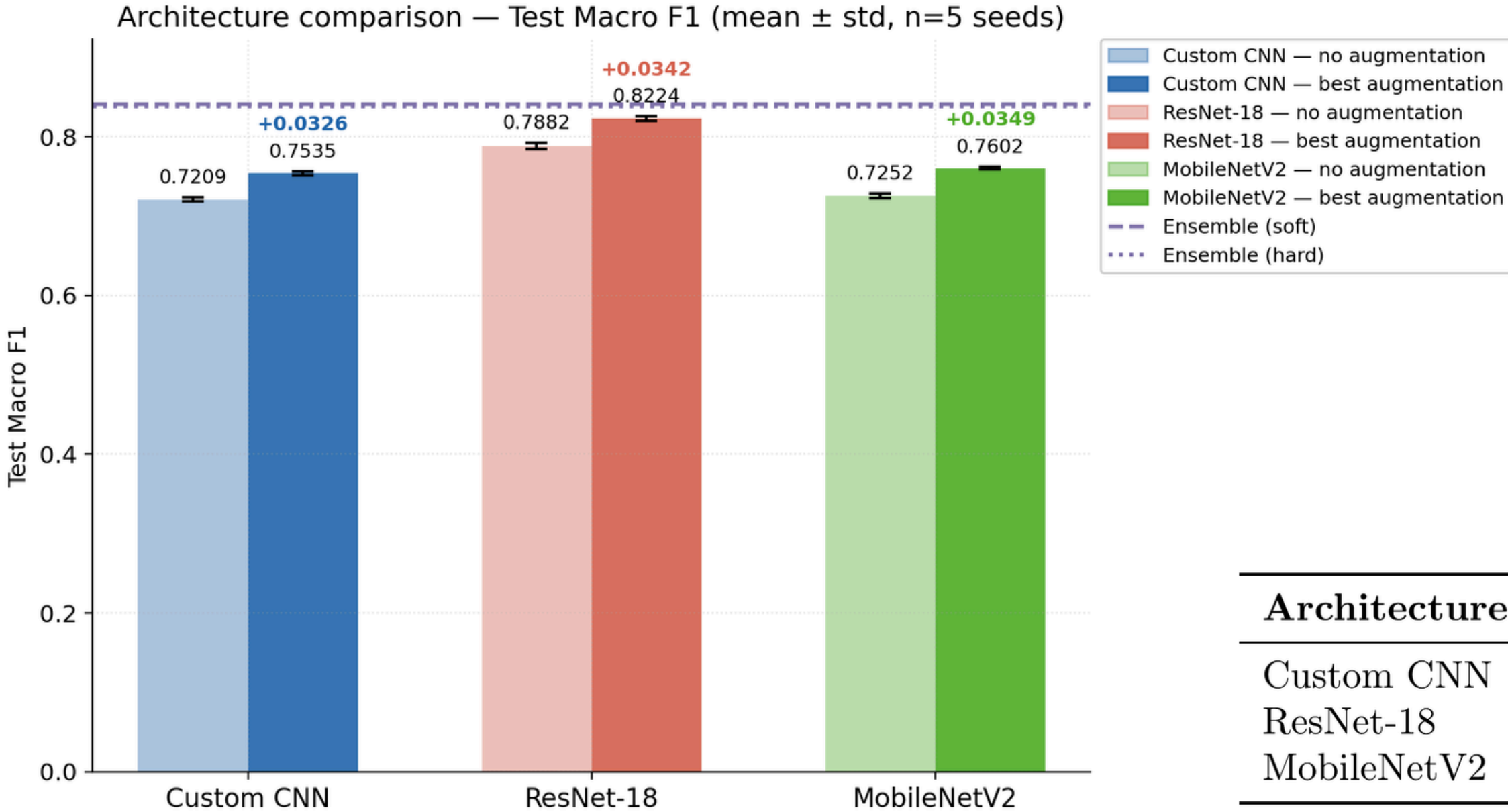
Model explainability - Integrated Gradients



Positive impact

Mean RGB

Architecture comparison & Ensembling



Method	Test Macro F1
Best single model (ResNet-18)	0.8224
Ensemble — hard voting	0.8382
Ensemble — soft voting	0.8412

Architecture	No aug. Test F1	Best aug. Test F1	Δ
Custom CNN	0.7214 ± 0.0025	0.7535 ± 0.0023	+0.032
ResNet-18	0.7991 ± 0.0059	0.8224 ± 0.0028	+0.034
MobileNetV2	0.7219 ± 0.0033	0.7602 ± 0.0016	+0.035



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